HW1 Report

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1. Model variants

I compared three text feature settings. First, Character TF-IDF divided each utterance into 2–5 character n-grams and assigned TF-IDF weights to capture informative local expressions without requiring Chinese word segmentation. Second, Character + Word TF-IDF added 1–3 word n-grams to capture word-level phrases, especially in English. Third, Character + Word TF-IDF with Bilingual Cues additionally included Chinese and English intent-related expressions, such as “price”, “warranty”, “別家”, and “太貴”, together with question marks, negation, utterance length, and language information. The extracted features in each setting were concatenated into a sparse numerical vector and provided to LinearSVC for classification. Since this was a multi-label task, I trained ten independent binary LinearSVC classifiers, one for each label. All three settings used the same 2,400/600 stratified train-validation split, C=1.7, class-balanced training, and a fixed decision threshold of 0.0. Therefore, the main difference among the three variants was their input feature representation.

Variant	Macro-F1	Micro-F1
Character TF-IDF	0.7309	0.7567
Character + Word TF-IDF	0.7102	0.7416
Character + Word + Bilingual Cues	0.7192	0.7481

Character-only TF-IDF performed best in this experiment. Character n-grams are suitable for Chinese because they do not require word segmentation and can directly capture expressions such as “太貴” and “別家”. Word features introduced additional sparsity, while bilingual cue features recovered part of the performance loss.

Initial mDeBERTa System

For the initial full system, I used microsoft/mdeberta-v3-base as the Transformer backbone. The model was fine-tuned for six epochs using two random seeds, 42 and 3407. I evaluated the complete system using five-fold out-of-fold validation.

The 3,000 training samples were divided into five folds. In each round, four folds were used for training and the remaining fold was used for validation. This was repeated five times so that every sample received a prediction from a model that had not been trained on that sample. These out-of-fold predictions were used to evaluate the system and select ensemble weights and thresholds without using test labels.

Final OOF System	Macro-F1	Micro-F1
Dual-seed mDeBERTa	0.8028	0.8124
mDeBERTa + LinearSVC Blend	0.8162	0.8240
Final Ensemble	0.8221	0.8306

Dual-seed mDeBERTa refers to two independently fine-tuned mDeBERTa models with random seeds 42 and 3407. Their output probabilities were averaged rather than their model weights. This reduces the variance caused by random initialization and training order.

For the **mDeBERTa + LinearSVC blend**, the probabilities from the two mDeBERTa models were first averaged. For each label, the averaged Transformer probability was then combined with the corresponding TF-IDF LinearSVC output using a label-specific weight selected from train-only OOF predictions. Therefore, this was a probability-level soft ensemble rather than a majority vote.

Based on the initial OOF analysis, Compare_Competitor, Doubt, and Too_Expensive were identified as relatively weak labels. I evaluated recent-context linear features for Compare_Competitor and Too_Expensive and a nested-OOF logistic stacker for Doubt. Chinese and English predictions were also calibrated separately, and the final decision thresholds were selected using only train-only OOF predictions. These additions increased the mDeBERTa system from 0.8162 to 0.8221 Macro-F1 and from 0.8240 to 0.8306 Micro-F1.

Final Backbone Selection

In the Chinese-versus-English backbone comparison described in Section 4, FacebookAI/xlm-roberta-base showed stronger validation performance than microsoft/mdeberta-v3-base. I therefore repeated the full training procedure with XLM-R to make a fair comparison.

The XLM-R experiment used the same 3,000 training samples, five-fold splits, random seeds 42 and 3407, six training epochs, dual-seed probability averaging, TF-IDF LinearSVC component, label-specific blending, and calibration procedure. The principal change was replacing the mDeBERTa backbone and tokenizer with XLM-R. All additional components were reevaluated using XLM-R OOF predictions and were retained only when they improved the validation results. For example, the nested logistic stacker was tested again but was not included in the final XLM-R inference system because it did not provide sufficient OOF improvement.

Final OOF System	Macro-F1	Micro-F1
mDeBERTa Final Ensemble	0.8221	0.8306
XLM-R Final Ensemble	**0.8358**	**0.8413**

Using the same train-only OOF evaluation, XLM-R improved Macro-F1 from 0.8221 to 0.8358 and Micro-F1 from 0.8306 to 0.8413. Therefore, the mDeBERTa results were retained for comparison, while XLM-R was selected as the final backbone. All model selection was based only on training-set OOF predictions without using test labels.

2. Does dialogue context help

I compared three input settings using the same train-only split as the lightweight model ablation: 2,400 training samples and 600 validation samples with seed 42. Among the validation samples, 539 contained dialogue context and 61 did not. All variants used the same character and word TF-IDF features and ten class-balanced LinearSVC classifiers with C=1.7 and a fixed decision threshold of 0.0.

The three settings were:

1. Utterance only: only the current customer utterance.

2. Full context: the current utterance and the available dialogue history, limited to the last 1,800 characters.

3. Recent context: the current utterance and the two most recent dialogue messages, limited to the last 1,200 characters.

Context was represented using a separate character TF-IDF vectorizer with 2–4 character n-grams and at most 80,000 features. The context feature values were multiplied by 0.35 to prevent the dialogue history from dominating the current utterance.

Input Setting	Macro-F1	Micro-F1
Utterance Only	0.7102	0.7416
Utterance + Full Context	0.7059	0.7387
Utterance + Recent Context	0.7078	0.7415

Adding context did not improve the overall performance in this lightweight experiment. Full context produced the lowest scores, while recent context was safer and achieved almost the same Micro-F1 as the utterance-only model.

Per-label results from the fixed-split LinearSVC ablation:

Label	Utterance Only	Recent Context	Difference
Ask_Price	0.8085	0.8021	-0.0064
Ask_Service	0.7638	0.7677	+0.0039
Ask_Spec	0.6899	0.6987	+0.0088
Compare_Competitor	0.5424	0.5263	-0.0161
Confirm_Order	0.9012	0.9012	0.0000
Doubt	0.6859	0.6860	+0.0002
Need_More_Evidence	0.8047	0.7937	-0.0110
Need_Time_To_Think	0.7516	0.7564	+0.0048
State_Need	0.7387	0.7387	0.0000
Too_Expensive	0.4151	0.4074	-0.0077

Recent context slightly improved Ask_Spec, Need_Time_To_Think, and Ask_Service but did not improve the overall scores. Full context performed worse because earlier turns introduced outdated intents and irrelevant keywords. Therefore, utterance-only features remained the default for LinearSVC.

In the initial mDeBERTa system, a context-aware LinearSVC selectively improved Compare_Competitor from 0.7062 to 0.7234 and Too_Expensive from 0.6704 to 0.6854. These are mDeBERTa results. After switching to XLM-R, context selection was repeated using OOF predictions.

3. Long inputs

I compared three strategies for handling long utterances. The experiment used the same fixed stratified 80/20 split as the previous ablations: 2,400 samples for training and 600 for validation with seed 42. Among the validation samples, 48 utterances contained more than 420 characters.

The complete dataset contained 208 utterances longer than 420 characters. After fixed stratified 80/20 split, 160 long utterances were included in the 2,400-sample training set, while 48 were included in the 600-sample validation set.

Truncation Strategies

1.Head-only: retain the first 420 characters.

2.Tail-only: retain the last 420 characters.

3.Head-tail: retain the first 140 and last 260 characters, with an omission marker between them.

Overall Results:

The results in this table were calculated only on the 48 long utterances in the validation set.

Long-input strategy	Macro-F1	Micro-F1
First 420 characters	0.5200	0.6396
Last 420 characters	0.5536	0.7041
First 140 + last 260	0.5690	0.6952

These results are from the fixed-split lightweight LinearSVC experiment rather than final OOF or unseen test-set evaluation. Because only 48 long validation samples were available, the per-label scores may have relatively high variance.

Single label result

Label	First 420	Last 420	Head-tail
Ask_Price	0.4444	0.2500	0.4444
Ask_Service	0.6471	0.8125	0.7805
Ask_Spec	0.7027	0.7222	0.6486
Compare_Competitor	0.4286	0.4286	0.5333
Confirm_Order	0.4000	0.6667	0.6667

Doubt	0.6512	0.7273	0.6829
Need_More_Evidence	0.7200	0.7692	0.7742
Need_Time_To_Think	0.7619	0.9091	0.9091
State_Need	0.0000	0.0000	0.0000
Too_Expensive	0.4444	0.2500	0.2500

Tail-only performed especially well for Ask_Service, Confirm_Order, Doubt, and Need_Time_To_Think, suggesting that the main intent often appears near the end of a long utterance. However, discarding the beginning reduced performance for Ask_Price and Too_Expensive.

Head-tail achieved the highest Macro-F1 and performed best for Compare_Competitor and Need_More_Evidence. Although tail-only achieved the highest Micro-F1, head-tail provided better balance across labels and preserved information from both ends of the utterance. Therefore, it was selected for the final pipeline. Long utterances are reduced to the first 140 and last 260 characters before being tokenized with a maximum sequence length of 512 tokens.

This long-input ablation was conducted while developing the initial mDeBERTa system. However, the reported scores were obtained from the lightweight TF-IDF–LinearSVC experiment rather than directly from mDeBERTa. The selected head-tail strategy was subsequently applied to the mDeBERTa pipeline and retained when the backbone was replaced with XLM-R.

4. Chinese vs. English

I compared two multilingual pretrained models: microsoft/mdeberta-v3-base and FacebookAI/xlm-roberta-base. The dataset contained 2,256 Chinese and 744 English samples. Both models used the same stratified 80/20 split with seed 42: 2,400 samples for training and 600 for validation. The validation set contained 453 Chinese and 147 English samples.

For a fair comparison, both models were fine-tuned for four epochs with the same dual-view input, consisting of a head-tail utterance view and a recent-context view. They used a maximum sequence length of 512, a backbone learning rate of 1.5×10^{-5}, a classification-head learning rate of 8×10^{-5}, and a fixed decision threshold of 0.5. Model-specific threshold tuning was not used.

Model / Language	Rows	Macro-F1	Micro-F1
mDeBERTa / Overall	600	0.7551	0.7684
mDeBERTa / Chinese	453	0.7563	0.7691
mDeBERTa / English	147	0.7502	0.7660
XLM-RoBERTa / Overall	600	0.7845	0.7977
XLM-RoBERTa / Chinese	453	0.7841	0.8005
XLM-RoBERTa / English	147	0.7859	0.7894

Chinese single label result

Label	mDeBERTa	XLM-R	Difference
Compare_Competitor	0.4595	0.6364	+0.1769
Ask_Spec	0.7470	0.7954	+0.0484
Doubt	0.6759	0.7178	+0.0419
Too_Expensive	0.6866	0.6154	−0.0712

English single label result

Label	mDeBERTa	XLM-R	Difference
Too_Expensive	0.4444	0.6250	+0.1806
Compare_Competitor	0.6154	0.7692	+0.1538
Need_More_Evidence	0.7302	0.7879	+0.0577
Need_Time_To_Think	0.8235	0.7647	−0.0588

For Chinese samples, XLM-RoBERTa improved Compare_Competitor from 0.4595 to 0.6364, Ask_Spec from 0.7470 to 0.7954, and Doubt from 0.6759 to 0.7178. However, mDeBERTa performed better on Chinese Too_Expensive, with 0.6866 compared with 0.6154. For English samples, XLM-RoBERTa improved Too_Expensive from 0.4444 to 0.6250, Compare_Competitor from 0.6154 to 0.7692, and Need_More_Evidence from 0.7302 to 0.7879. In contrast, mDeBERTa performed better on English Need_Time_To_Think, achieving 0.8235 compared with 0.7647. Because the English subset contained only 147 samples, differences for rare labels may have higher variance.

Since XLM-RoBERTa performed better in this controlled comparison, I subsequently retrained the complete pipeline using XLM-R with the same five-fold OOF procedure, seeds 42 and 3407, six epochs, LinearSVC blending, context selection, and calibration. The full XLM-R system achieved 0.8358 Macro-F1 and 0.8413 Micro-F1, outperforming the original mDeBERTa system, which achieved 0.8221 and 0.8306. Therefore, XLM-RoBERTa was selected as

the backbone of the final submitted system. All comparisons were based only on train-only validation or OOF predictions.

5. Additional Exploration

(1) Five-fold Out-of-Fold Validation

Instead of selecting ensemble parameters from a single validation split, I used five-fold out-of-fold (OOF) validation on all 3,000 training samples. In each fold, 2,400 samples were used for training and the remaining 600 were used for prediction. This was repeated five times, so every training sample received a prediction from a model that had not been trained on that sample.

All ensemble weights, decision thresholds, and component-selection decisions were based only on these OOF predictions. No test labels were used. This provides more stable evidence than a single split, although the resulting scores are still train-only validation results.

(2) Dual-seed Transformer Ensemble

I independently fine-tuned mDeBERTa using random seeds 42 and 3407. For each sample and label, the two output probabilities were averaged:

$$p_l^T = \frac{p_{l,42} + p_{l,3407}}{2}$$

The model parameters themselves were not average. Probability averaging reduces variance caused by random initialization, data order, and optimization. The same probability-averaging procedure was used for XLM-R.

(3) Per-label Transformer and LinearSVC Blending

I trained a second model family using Character TF-IDF, Word TF-IDF, bilingual cue features, and ten class-balanced LinearSVC classifiers. Character n-grams used lengths 1–5, word n-grams used lengths 1–3, and the maximum feature counts were 240,000 and 120,000, respectively.

The LinearSVC regularization parameter was selected independently for each label from:

$$C \in \{0.6, 1.2, 2.4, 4.0\}$$

The LinearSVC decision values were transformed using the sigmoid function and combined with the average Transformer probability:

$$p_l = \alpha_l p_l^T + (1 - \alpha_l) p_l^L$$

The Transformer weight α_l was searched from 0 to 1 in steps of 0.05. A separate threshold was also selected for each label. This allowed some labels to rely more on semantic Transformer features and others to use more exact lexical evidence from TF-IDF.

OOF system	Macro-F1	Micro-F1
LinearSVC only	0.7600	0.7706
Dual-seed mDeBERTa	0.8028	0.8124
mDeBERTa + LinearSVC blend	0.8162	0.8240
Dual-seed XLM-R	0.8196	0.8257
XLM-R + LinearSVC blend	0.8308	0.8356

After the backbone comparison, I applied the same per-label blending method to XLM-R. The same LinearSVC models, label-specific C selection, Transformer-weight search, and threshold-selection procedure were retained; only the Transformer probability source was changed from mDeBERTa to XLM-R. Blending improved the dual-seed XLM-R model from 0.8196 to 0.8308 Macro-F1 and from 0.8257 to 0.8356 Micro-F1. This confirms that TF-IDF lexical features remained complementary even when the stronger XLM-R backbone was used.

(4) Weak-label Context Models

The following context-model results were obtained from the initial mDeBERTa system.

The plain blend showed that Compare_Competitor, Doubt, and Too_Expensive remained the weakest labels. I therefore trained an additional context-aware LinearSVC only for these labels.

Its input contained: The current customer utterance、the two most recent dialogue messages、at most 1,200 characters of recent context、explicit markers distinguishing the current utterance from dialogue history

I searched for a three-source combination:

$$p_l = w_T p_l^T + w_L p_l^L + w_C p_l^C$$

where p_l^C is the context-aware LinearSVC score, and the three weights sum to one. Chinese and English thresholds were selected separately. A component was retained only if its OOF F1 improvement exceeded 0.002.

The selected context weights were:

Label	Transformer	Plain LinearSVC	Context LinearSVC
Compare_Competitor	0.200	0.675	0.125
Too_Expensive	0.175	0.375	0.450

Result:

Label	Plain blend	Context blend	Improvement
Compare_Competitor	0.7062	0.7234	+0.0172
Too_Expensive	0.6704	0.6854	+0.0150

The relatively large context weight for Too_Expensive indicates that price objections sometimes depend on information from the preceding dialogue. However, context was not added to every label because it could also reintroduce outdated intents.

The weak-label context experiment was repeated after replacing mDeBERTa with XLM-R. The context-aware LinearSVC was selectively retained for Doubt and Too_Expensive, improving the overall OOF Macro-F1 from 0.8308 to 0.8330 and Micro-F1 from 0.8356 to 0.8383.

(5) Nested-OOF Logistic Stacker for Doubt

This stacker result also belongs to the initial mDeBERTa experiments.

A simple weighted average may not model complex disagreement between the Transformer and LinearSVC. Therefore, tested a logistic-regression meta-classifier for the three weak labels.

Its inputs included: Transformer probabilities for all labels、LinearSVC scores for all labels、logit-transformed scores、products between model scores、absolute disagreement between the two models、observable bilingual cue features

The logistic-regression C value was searched over:

$$\{0.01,0.03,0.1,0.3,1,3,10\}$$

Both balanced and unweight variants were evaluated. Nested five-fold predictions were used so that the meta-classifier was not evaluated on samples used to train that meta-classifier.

Only Doubt exceeded the minimum improvement requirement:

Label	Plain blend	Stacker	Decision
Compare_Competitor	0.7062	0.7067	Rejected
Doubt	0.7337	0.7399	Retained
Too_Expensive	0.6704	0.6667	Rejected

For Doubt, the selected logistic regression used $C = 0.1$, no class weighting, and an initial threshold of 0.39. The stacker improved Doubt because uncertainty is often better identified from the disagreement between semantic and lexical models than from either model alone.

The nested-OOF logistic stacker was also re-evaluated using XLM-R predictions. However, it did not provide sufficient OOF improvement and was therefore excluded from the final XLM-R system.

Full-Pipeline Backbone Comparison

The context-model and nested-stacker experiments above were conducted using the initial mDeBERTa system. After the controlled Chinese-versus-English experiment showed that XLM-R was stronger, I repeated the complete training and model-selection procedure with XLM-R. Both backbones used the same five-fold OOF protocol, seeds 42 and 3407, six training epochs, LinearSVC blending, and OOF-based component selection and calibration.

After final calibration, the three main weak labels obtained:

OOF Configuration	**mDeBERTa Macro-F1**	**mDeBERTa Micro-F1**	**XLM-R Macro-F1**	**XLM-R Micro-F1**
Dual-seed Transformer	0.8028	0.8124	**0.8196**	**0.8257**
Transformer + LinearSVC	0.8162	0.8240	**0.8308**	**0.8356**

Final selected ensemble	0.8221	0.8306	**0.8358**	**0.8413**

XLM-R outperformed mDeBERTa at every comparable stage. In the final ensemble, it improved Macro-F1 by 0.0137 and Micro-F1 by 0.0107. The same candidate enhancements were evaluated again using XLM-R OOF predictions. Therefore, XLM-R was selected as the backbone of the final submitted model.

Evaluation Protocol

All reported F1 scores were obtained from training-data validation. Controlled ablations used a fixed 600-sample validation split, while the ensemble comparisons used five-fold OOF predictions covering all 3,000 training samples. The official test set was used only to generate submission predictions; its labels were not used for training, model selection, threshold tuning, or F1 calculation.